Muhammad Osama Zeeshan

Computer Vision Scientist | Deep Learning Researcher | Software Developer
Montréal, QC, Canada • +1-438-357-6482 • osamaz.oz31@gmail.com
• osamazeeshan.github.io • LinkedIn • Google Scholar

PROFESSIONAL SUMMARY

Computer Vision Scientist completing a PhD in June 2026, with experience developing deep learning models for real-world visual understanding, multimodal learning, and robust model adaptation. Skilled in building ML pipelines using Python, PyTorch, TensorFlow, and OpenCV, with experience working on noisy, heterogeneous visual datasets and systematically improving model performance through evaluation, ablation studies, and data preprocessing. Published in ICLR, WACV, and CVPR Workshops, with prior industry experience developing production software systems and applied computer vision pipelines.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, SQL

Frameworks & Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, SciPy, OpenCV

Computer Vision: Image Classification, Object Detection, Localization, Tracking, OCR, Image Processing, Feature Extraction, OpenCV

Data & ML Systems: Data Preprocessing, Data Augmentation, Noisy Data Handling, Large-scale Experiments, Model Benchmarking, Reproducible ML Pipelines

Machine Learning: Deep Learning, CNNs, Transformers, Multimodal Learning, Domain Adaptation, Model Evaluation, Classical ML

Software Engineering: RESTful APIs, Backend Systems, Modular Design, Debugging, Git, Linux

EDUCATION

PhD in Systems Engineering | École de technologie supérieure (ÉTS) Jan 2022 – June 2026 (expected)
Focus: Deep Learning, Multimodal AI, Domain Adaptation, Human Behavior Analysis

MSc in Data Sciences | Bahria University Sep 2018 – Dec 2020

BSc in Computer Sciences | Air University Sep 2010 – May 2014

RESEARCH EXPERIENCE

Computer Vision Researcher | LIVIA & ILLS Labs – ÉTS, Montreal, Canada Jan 2022 – Present

- Develop deep learning models for visual behavior and expression analysis using Python and PyTorch.
- Design personalized and domain-adaptive computer vision methods to improve robustness across subjects, environments, and data conditions.
- Propose multimodal learning approaches integrating visual, physiological, and audio signals for robust prediction in real-world settings.
- Build end-to-end ML pipelines for data preprocessing, training, evaluation, ablation studies, and reproducible experimentation.
- Conduct large-scale experiments to evaluate accuracy, robustness, generalization, and cross-domain performance on noisy real-world datasets.

- Co-developed the BAH dataset for ambivalence and hesitancy recognition, including benchmarking and evaluation protocols.
- Collaborate with interdisciplinary researchers to refine model design, evaluation criteria, and experimental methodology.

Visiting Research Scientist | STARTS, Inria – French National Institute for Research in Digital Science and Technology, Valbonne, France, Jan 2026 – Apr 2026

- Conducted collaborative research on computationally efficient ML methods for personalized human behavior and expression analysis.
- Investigated Test-Time Multimodal Domain Adaptation approaches, eliminating the need for original training data while improving model robustness and efficiency.
- Performed a technological watch on emerging AI/ML research relevant to efficient personalized modeling.

INDUSTRY EXPERIENCE

Senior Software Engineer | Jin Technologies Jan 2019 – Jan 2022

- Built scalable backend systems and RESTful APIs for production data-driven applications.
- Designed modular, maintainable software components with focus on reliability, performance, and scalability.
- Collaborated with cross-functional teams to translate product requirements into production-ready systems.
- Applied software engineering best practices, including debugging, documentation, version control, and system optimization.

Software Developer – Video Content Retrieval | Bahria University Research Lab Sep 2017 – Jan 2019

- Developed computer vision pipelines for Urdu text detection and recognition in low-quality video streams.
- Built indexing and retrieval systems for large multimedia datasets, enabling visual content search.
- Designed data preprocessing workflows for noisy real-world video data, including low-resolution frames and variable lighting conditions.
- Applied Python, OpenCV, and TensorFlow to build and evaluate applied image/video processing models.

Android Developer | Steprobotics Feb 2015 – Aug 2017

- Developed computer vision algorithms for skyline and shading analysis to support solar irradiance estimation.
- Built the Step Solar Android application, integrating geospatial analysis, mobile data capture, and external APIs.
- Designed image-processing workflows for real-world outdoor visual data with variable lighting and scene conditions.
- Collaborated with engineering teams to integrate CV outputs into a user-facing mobile application.

RESEARCH IMPACT & PUBLICATIONS

- Published in top-tier AI venues: ICLR, WACV, and CVPR Workshops.
- 300+ citations, h-index: 8 (Google Scholar).
- Co-developed the BAH dataset — a publicly available benchmark for ambivalence and hesitancy recognition in behavioral analysis.

AWARDS & HONORS

- FRQNT Award (2025–2026): "Personalized Deep Learning Models for Diverse Individuals", Quebec, Canada.
- 4-Year PhD Scholarship (2022–2026): Fully funded research position at ÉTS, Montréal.
- Top 100 – Hello Tomorrow Global Summit, Le Centquatre, Paris, France (2019).